

OIS-002 — Implementation Architecture

Status: Foundational

Authority: Implementation

Depends on:

- OIS-000
 - OIS-001
 - OAS-000 through OAS-031
 - OCS-000 through OCS-032
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Purpose

This specification defines the canonical implementation architecture of Organodynamics.

The Constitutional Specification defines immutable principles.

The Architectural Specification defines stable organization.

This specification defines how that organization is expressed as executable systems.

Implementation Architecture SHALL preserve architectural intent while allowing unlimited engineering diversity.

1. Implementation Objective

The objective of the Implementation Architecture is to separate constitutional behavior from implementation mechanics.

Implementation SHALL organize execution so that constitutional evolution remains independent of technological evolution.

Every executable component SHALL exist only to realize architectural responsibilities.

2. Canonical Layers

Every conforming implementation SHALL distinguish between the following implementation layers.

Constitution (OCS)

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Architecture (OAS)

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Application Services

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Execution Engine

↓

Persistence

↓

Infrastructure

Each layer SHALL depend only upon the layers above it.

No lower layer SHALL redefine higher-layer behavior.

3. Constitutional Boundary

The Constitutional layer SHALL never execute implementation logic.

Its responsibility is normative definition.

Implementation SHALL consume constitutional definitions.

It SHALL never modify them.

4. Architectural Boundary

The Architectural layer SHALL organize responsibilities.

It SHALL define:

Interfaces.

Patterns.

Relationships.

Architectural behavior.

Implementation SHALL realize these responsibilities.

It SHALL not reinterpret them.

5. Application Services

Application Services SHALL coordinate constitutional operations.

Typical responsibilities include:

Identity resolution.

Conversation orchestration.

Decision execution.

Library interaction.

Federation coordination.

Application Services SHALL remain stateless whenever constitutionally possible.

6. Execution Engine

The Execution Engine SHALL realize operational behavior.

Responsibilities MAY include:

Event processing.

Workflow execution.

Scheduling.

Rule evaluation.

Concurrency.

Recovery.

Execution SHALL remain replaceable.

7. Persistence Layer

Persistence SHALL preserve constitutional information.

Persistence SHALL remain responsible only for:

Durability.

Retrieval.

Consistency.

Recovery.

Persistence SHALL never determine constitutional meaning.

8. Infrastructure Layer

Infrastructure SHALL provide operational capabilities.

Examples include:

Networking.

Storage engines.

Cloud platforms.

Operating systems.

Monitoring.

Deployment.

Infrastructure SHALL remain constitutionally invisible.

9. Dependency Direction

Implementation dependencies SHALL always point toward greater constitutional stability.

Infrastructure SHALL depend upon Persistence.

Persistence SHALL depend upon Execution.

Execution SHALL depend upon Application Services.

Application Services SHALL depend upon Architecture.

Architecture SHALL depend upon the Constitution.

Reverse dependencies SHALL be prohibited.

10. Replaceability

Every implementation layer SHALL be independently replaceable.

Replacement SHALL preserve:

Behavior.

Identity.

Historical reconstruction.

Architectural compatibility.

Constitutional equivalence.

11. Verification

Verification SHALL demonstrate:

Correct dependency direction.

Architectural compliance.

Behavioral equivalence.

Operational explainability.

Independent replaceability.

12. Implementation Architecture Test

Implementation Architecture satisfies this specification when two engineering teams, using completely different implementation technologies, independently construct systems whose observable constitutional behavior remains equivalent despite differences in internal structure.

Conformance

An implementation conforms to this specification when every execution layer preserves constitutional meaning, architectural intent, behavioral equivalence, and implementation replaceability independently of technology choices.

Implementation Principle

Software inevitably becomes obsolete.

Infrastructure inevitably changes.

Execution models inevitably evolve.

A correct implementation therefore organizes change so that the only permanent dependency is upward—
toward architecture,

and ultimately toward the Constitution.

Everything else must remain free to be rebuilt.

End of Specification OIS-002